

Databases

Course 9

Oracle SQL Functions

SQL-Oracle

Retrieving data

Statement	Description
SELECT	Retrieves data from one or more tables

SELECT clauses	Description
WHERE <i>condition</i>	retrieves only those records that fulfill a specified condition.
start with .. connect by	select data that has a hierarchical relationship (manager->employee or thing->parts)
GROUP BY <i>columns, single row functions</i>	grouping lines to identify the common value of groups (aggregate values calculated by group functions)
HAVING <i>condition, agg functions</i>	restricts groups created by the GROUP BY clause based on a group function condition
ORDER BY <i>columns</i>	Orders the rows (implicitly ascending)

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Single row functions

String manipulation

Syntax	Result
UPPER (s) / LOWER (s)	string
INITCAP (s)	string
CONCAT (s1,s2)	string
LPAD (s1,n,s2) / RPAD (s1,n,s2)	string
LTRIM (s1,s2) / RTRIM (s1,s2)	string
TRIM([[LEADING TRAILING BOTH] trim_character FROM] string1)	string
LENGTH (s)	number
INSTR (s1,s2, poz,n)	number
SUBSTR (s,poz,n)	string
REPLACE (s1,s2,s3)	string

Ex:

- `select LPAD('DBMS',3), LPAD('DBMS',5), LPAD('DBMS',9,'01') FROM DUAL;`
- `select TRIM(TRAILING '1' FROM '11DB111'), TRIM('1' FROM '11DB111'), TRIM('1' FROM '12DB121'), TRIM(' D B ') FROM DUAL;`
- `SELECT product_name, LTRIM(product_name, 'Monitor ') "Short Name" FROM product_information WHERE product_name LIKE 'Monitor%';`
- `select INSTR('dbmsms','ms',-1,1), INSTR('dbmsms','ms',1,1) from dual;`
- `SELECT REPLACE('DBMS_DB','DB'), REPLACE('DBMS_DB','DB','?') from dual;`

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Manipulates numbers or dates

Sintax	Result
ROUND (n,i)	Number/Date
TRUNC (n,i)	Number/Date
MOD (n1,n2)	number

Ex:

- `select round(145.78,1), round(145.78,-1), round(145.78) from dual;`
- `select trunc(145.78,1), trunc(145.78,-1), trunc(145.78) from dual;`
- `select mod(4,2), mod(11,2) from dual;`
- `select round (sysdate,'Year'), trunc (sysdate,'Year') from dual;`
- `select round (sysdate,'Q'), trunc (sysdate,'Q') from dual;`
- `select round (sysdate,'Month'), trunc (sysdate,'Month') from dual;`
- `select round (sysdate,'Day'), trunc (sysdate,'Day') from dual;`
- `select round (sysdate,'DD'), trunc (sysdate,'DD') from dual;`

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Functions than manipulate dates

Sintax	Result
SYSDATE	Date
NEXT_DAY (d,s)	Date
LAST_DAY (d)	Date
ADD_MONTHS (d,n)	Date
MONTHS_BETWEEN (d1,d2)	Number
EXTRACT (DAY FROM d) EXTRACT (MONTH FROM d) EXTRACT (YEAR FROM d)	Number

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SQL special functions

Syntax

NVL (*e1,e2*)

NULLIF (*e1,e2*)

DECODE (*e,search,result,....,default*)

CASE *expr* WHEN *cond* THEN *rez* ... ELSE *rez* END (CASE is a SQL expression that works similar to a single row function)

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Conversion functions

Syntax

TO_NUMBER(*s,format*)

TO_CHAR (*n,format*)

TO_CHAR (*d,format*)

TO_DATE (*s,format*)

TO_TIMESTAMP (*s,format*)

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Aggregate functions

Syntax
SUM(<i>e</i>)
AVG(<i>e</i>)
COUNT(<i>e</i>)
MIN(<i>e</i>)
MAX(<i>e</i>)

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Aggregate functions (cont)

APPROX_COUNT_DISTINCT
AVG
COLLECT
CORR
CORR_*
COUNT
COVAR_POP
COVAR_SAMP
CUME_DIST
DENSE_RANK
FIRST
GROUP_ID
GROUPING
GROUPING_ID
LAST
LISTAGG
MAX
MEDIAN
MIN
PERCENT_RANK
PERCENTILE_CONT
PERCENTILE_DISC
RANK
REGR_ (Linear Regression) Functions
STATS_BINOMIAL_TEST
STATS_CROSSTAB
STATS_F_TEST
STATS_KS_TEST
STATS_MODE
STATS_MW_TEST
STATS_ONE_WAY_ANOVA
STATS_T_TEST_*
STATS_WSR_TEST
STDDEV
STDDEV_POP
STDDEV_SAMP
SUM
SYS_OP_ZONE_ID
SYS_XMLAGG
VAR_POP
VAR_SAMP
VARIANCE
XMLAGG

Group by

- SELECT *column_name(s)*
FROM *table_name*
WHERE *condition (not on aggregate functions)*
GROUP BY *column_name(s)*
HAVING *condition (usually on aggregate functions)*
ORDER BY *column_name(s)*;

Group by examples

- SELECT COUNT(customer_id),
country FROM customers
GROUP BY country
ORDER BY COUNT(customer_id)
DESC;

- SELECT COUNT(customer_id),
country FROM customers
GROUP BY country
HAVING AVG(credit_limit)>200000
ORDER BY count(customer_id) DESC;